A counterexample to conjecture 00000000650

quantumtool

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Statement and counterexample

The conjecture asserts the bound

$$b(K) \geq 2 + \left\lceil \frac{g_3(K)}{b(K) - 1} \right\rceil$$

and proposes the floor version

$$b(K) \geq 2 + \left\lfloor \frac{g_3(K)}{\lfloor b(K) - 1 \rfloor} \right\rfloor,$$

with equality on every $(2, n)$-torus knot. Take the trefoil $K = T(2, 3)$. The bridge number of a torus knot $T(p, q)$ is $\min(p, q)$, and its Seifert genus is $(p - 1)(q - 1)/2$. Thus

$$b(T(2, 3)) = 2, \qquad g_3(T(2, 3)) = 1.$$

Both displayed right-hand sides are therefore $2 + 1 = 3$, while the left-hand side is 2. Both inequalities fail.

For completeness, the two invariant values can also be read directly from standard facts about the trefoil. A 2-bridge presentation gives $b(K) \leq 2$, while $b(K) = 1$ characterizes the unknot and the trefoil is nontrivial, so $b(K) = 2$. The Seifert algorithm on its positive three-crossing diagram gives a genus-one Seifert surface. Its Alexander polynomial has span two, so the Alexander-polynomial genus bound gives $g_3(K) \geq 1$; hence $g_3(K) = 1$.

The displayed floor denominator is $\lfloor 2 - 1 \rfloor = 1$, so the floor version fails at the same knot. The issue is already present in the first inequality; replacing a ceiling by a floor does not repair it.

Lean verification

The Lean 4 project records the standard formulas for torus-knot bridge number and genus, evaluates them at $(2, 3)$, and proves that each natural-number inequality is false. The arithmetic theorem has no additional axioms. The topological identification of the trefoil invariants is justified above.